

Homework for Chap 11

Which trend would you expect for the zone boundary LA and TA phonons when going from ZnO via ZnS and ZnSe to ZnTe. Compare your thought with data in the literature. What do you expect for zone center optic phonons?

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① $\text{ZnO} \rightarrow \text{ZnS} \rightarrow \text{ZnSe} \rightarrow \text{ZnTe} \Rightarrow$ zone boundary LA/TA phonon trend: 无 LA/TA, 从 $\text{ZnO} \rightarrow \dots \rightarrow \text{ZnTe}$ 频率逐渐下降

$\Rightarrow$ phonon frequency: $\omega \propto \sqrt{\frac{k}{\mu}}$

约化质量 $\mu \uparrow$, 键力常数 $k \downarrow \Rightarrow \omega \downarrow$
键弱, 晶格常数大

② ZnO 纤锌矿结构, 键很强 $\rightarrow \omega_{\max}$

$\omega: \text{ZnO} > \text{ZnS} > \text{ZnSe} > \text{ZnTe} \Leftarrow$

$\text{ZnS}: 80-120 \text{ cm}^{-1}$
$\text{ZnSe}: 50-100 \text{ cm}^{-1}$
$\text{ZnTe}: < 80 \text{ cm}^{-1}$

② zone center optic phonons

其频率同样 $\text{ZnO} \rightarrow \dots \rightarrow \text{ZnTe}$ 下降

material	TO cm^{-1}	LO cm^{-1}
ZnO	~ 380	~ 574
ZnS	~ 270	~ 350
ZnSe	~ 216	~ 253
ZnTe	~ 178	~ 208

Raman/IR

光学声子随阳离子质量增加呈显著下降

$\Rightarrow \omega: \text{ZnO} \rightarrow \dots \rightarrow \text{ZnTe}$ 逐渐下降

原因: 1. $\mu \uparrow$ $\text{O} \rightarrow \dots \rightarrow \text{Te}$

2. $k \downarrow$ 键逐渐变弱